

$\neg(P(a))$ is False. By double negation law, $P(a)$ is False for all elements a in the domain. $\therefore$ For each a in the domain, by the defn. of OR, $P(a) \vee A$ is False. $\therefore$ By the defn. of ~~universal~~ universal quantifier, $\forall x(P(x) \vee A)$ is False. $\therefore$ From the defn. of existential quantifier, $\exists x(P(x) \vee A)$ is False.

$\therefore (\exists x P(x)) \vee A$ is False $\Rightarrow \exists x(P(x) \vee A)$ is False

Let $\exists x(P(x) \vee A)$ is False. By the defn. of negation, $\neg \exists x(P(x) \vee A)$ is True. By De Morgan's law of quantifiers, $\forall x \neg(P(x) \vee A)$ is True. By 2nd De Morgan's law, $\forall x(\neg P(x) \wedge \neg A)$ is True. By defn. of universal quantifier, for each element a in the domain, $\neg P(a) \wedge \neg A$ is True. By the defn. of AND, $\neg P(a)$ is True and $\neg A$ is True. By the defn. of negation, $\neg(\neg P(a))$ is False and $\neg(\neg A)$ is False. By the double-negation law, $P(a)$ is False and A is False. $\therefore$ For all elements a in the domain, $P(a)$ is False. By defn. of existential quantifier, $(\exists x P(x))$ is False by necessity since by the defn. of universal quantifier, $(\forall x P(x))$ is False. By the defn. of OR,

$(\exists x P(x)) \vee A$ is False.

$\therefore \exists x(P(x) \vee A)$ is False $\Rightarrow (\exists x P(x)) \vee A$ is False

$\therefore$ They are logically equivalent $\square$

49) Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

$(\forall x P(x)) \wedge A \equiv \forall x(P(x) \wedge A)$

Let $(\forall x P(x)) \wedge A$ be True. By defn. of AND, $(\forall x P(x))$ is True and A is True. By defn. of universal quantifier, for all elements a in the domain, $P(a)$ is True. By defn. of AND, $P(a) \wedge A$ is True for all elements a in the domain. By defn. of universal quantifier, $\forall x(P(x) \wedge A)$ is True.